

Optimization of Gauss Method to describe with most accuracy the orbits of Near Earth Asteroids - NEAs

Johana Murcia¹, Nicolás Molina², Miguel E. Gámez López³,
Néstor Méndez¹, Eduardo Mafla⁴ and Camilo Delgado-Correal⁵ 

¹Department physics, National Pedagogical University

²Astronomical observatory, National University of Colombia

³Department physics, National University of Colombia

⁴Educational institution Compartir of Mosquera

⁵Faculty of Engineering, Distrital University FJC
email: mcdelgadoc@udistrital.edu.co

Abstract. In this work we reviewed the Gauss method to infer the orbits of minor bodies of the solar system, as the identification of optimization parameters to infer the orbital elements of two asteroids near to the Earth (NEAs): 5587 (1990 SB) and 4953 (1990 MU)), already cataloged and named by the Minor Planet Center - MPC. We used the database of JPL - Horizons and also included an analysis using data of Gaia Data Release 3 (DR3). We did an statistic analysis between distribution of points correspondent to orbital elements that we obtained and the inferred by the Jet Propulsion Laboratory-JPL. When data is crossed with the orbital parameters of Small-Body Database (JPL), we analyzed the deficiency grades of the method by statistic comparation with state vector method between orbital elements inferred with the implemented method and the accepted by the community, obtaining a relative error for the orbits calculated of 0.424149% and 0.416237% for the body 5587 (1990SB), and 0.257968% and 0.223521% for the body 4953 (1990MU). The first error value mentioned corresponds to an orbit calculated with the database JPL- Horizons, and the second value to an orbit calculated with the database Gaia (DR3), for each body in study. In the first phases of implementation of the code, it was found that the restrictions of the traditional method are overcome under the additional parameters that are proposed, resulting in orbits that are better approximated than those determined by NASA Team at the time of observation corresponding to the collection of data for the different bodies analyzed in the framework of this work. The best approximations are established with respect to the calculation of the orbital elements through the numerical solutions incorporated in the NASA SPICE kernel in python for a period of 100 years, with a step of 1 month. We found that our data are quite close to the curves that represent the variations of each of the 5 orbital elements involved in our analysis, namely: a, e, i, ω, Ω . Finally, it should be noted that our method could contribute to the estimation of orbits for minor bodies of the Solar System from observational data, which could easily be taken by using small telescopes. Thus, it would enrich processes that seek to expand the coverage of observatories focused on estimating the orbits of minor bodies in the solar system.

Keywords. Method Gauss, Asteroids, Optimization parameters

1. Introduction

A major problem in orbit determination is the calculation of the orbital elements of a celestial body using a given set of angular observations(α_i, δ_i)([Gronchi \(2004\)](#)), for this,

the Gaussian method is implemented (Gurfil & Seidelmann (2016)). The focus of the problem in this work is to find the distance from the Sun of two near-Earth asteroids (5587 (1990 SB) and 4953 (1990 MU)), knowing only their relative positions in the celestial vault to three observation times. For this reason, a possible optimization of the Gaussian method is proposed, taking into account **the interval of separation of observation dates**, because by setting this criterion well, a well-defined orbit can be obtained.

2. Gauss method for preliminary orbit inference

The Gauss's method for determining preliminary orbits is an iterative process and uses vector algebra for all his deductions.

We can see that between each set of vectors there is the following mathematical relationship: $\vec{r}_i = -\vec{R}_i + \vec{\rho}_i$, where $\vec{r}_i$ are the Asteroid position vectors with respect to the Sun, $\vec{R}_i$ represent the position of the Earth with respect to the sun and $\vec{\rho}_i$ is the position of the smaller body with respect to the Earth, with $i = 1, 2, 3$ corresponding to the indices of the radio vectors of each observation.

Notice that $\vec{\rho}_i$ is a geocentric vector, and since the observations are referred to the celestial sphere, this vector can be rewritten in terms of its components using spherical coordinates, thus:

$$\vec{\rho}_i = \rho_i (\cos \alpha_i \cos \delta_i \hat{i} + \sin \alpha_i \cos \delta_i \hat{j} + \sin \delta_i \hat{k}) = \rho_i \hat{u}_i \quad (2.1)$$

On the same hand, the following vector sum can be established between each of the vectors $\vec{r}_1$, $\vec{r}_2$ and $\vec{r}_3$ in the figure (1) given their coplanarity relationship.

$$\vec{r}_2 = c_1 \vec{r}_1 + c_3 \vec{r}_3 \quad (2.2)$$

Substituting the equation (2.1) on the equation (2.2) and organizing the expression, we obtained:

$$\rho_2 \hat{u}_2 - c_1 \rho_1 \hat{u}_1 - c_3 \rho_3 \hat{u}_3 = \vec{R}_2 - c_1 \vec{R}_1 - c_3 \vec{R}_3 \quad (2.3)$$

Now the key point to Gauss's method is to find some value of ρ_i , in terms of what is already known, so we must eliminate ρ_i like ρ_1 and ρ_3 and find ρ_2 . To do that, a vector must be found that is perpendicular to $\hat{u}_1$ and $\hat{u}_3$, this is the vector $(\hat{u}_1 \times \hat{u}_3)$, developing a vector elimination method and applying the corresponding algebra, the following general expression can be deduced:

$$\rho_j = \frac{-\vec{R}_j \cdot (\hat{u}_i \times \hat{u}_k) + c_i \vec{R}_i \cdot (\hat{u}_i \times \hat{u}_k) + c_k \vec{R}_k \cdot (\hat{u}_i \times \hat{u}_k)}{c_j \hat{u}_j \cdot (\hat{u}_i \times \hat{u}_k)} \quad (2.4)$$

With i, j, k different swapping between 1,2,3. The terms c_1 and c_3 that result from the equation (2.4), are obtained by means of the Lagrange coefficients which result from considering a vector function that can be expressed as a Taylor-type approximation to express a position vector which results from the analysis of two bodies between the Sun and the minor body in its second position.

$$c_1 = \frac{g_3}{f_1 g_3 - g_1 f_3} \quad c_3 = \frac{-g_1}{f_1 g_3 - g_1 f_3}$$

Initially, the method assumes knowing the Lagrange coefficients f_i and g_i , that depend on $\vec{r}_2$ and its time derivative. Any value of $\vec{r}_2$ can be assumed, which will be progressively updated as the procedure iterates again, hoping to reach a convergence value of $\vec{r}_2$ given

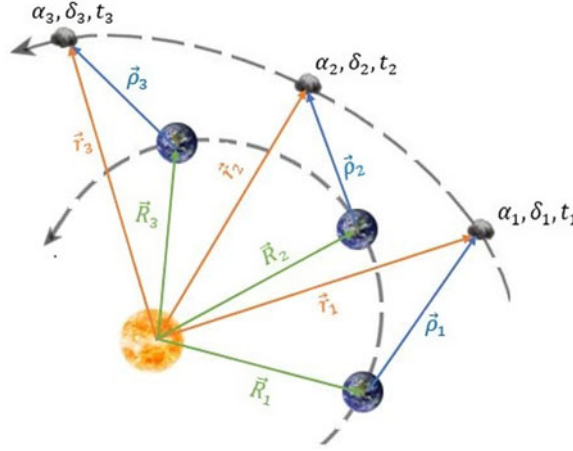


Figure 1. Graphical representation of the Sun-Earth-Asteroid position vectors. Image taken and modified from [Zuluaga \(2021\)](#)

a certain number of iterations, knowing ρ_2 it is possible to find the real value of $\vec{r}_2$, applying the following equation:

$$\vec{r}_2 = \rho_2 \hat{u}_2 - \vec{R}_2 \quad (2.5)$$

Generally with $(i = 1, 2, 3)$. Once the position of the body has been determined with the equation (2.5), to obtain the orbital elements, it is necessary to know the speed of the smaller body, for this the following equation is used [Curtis \(2013\)](#):

$$\vec{r}_2 = \frac{f_1 \vec{r}_3 - f_3 \vec{r}_1}{f_1 g_3 - f_3 g_1} \quad (2.6)$$

Finally, with the state vector that describes the movement of the asteroid, the orbital elements are calculated as have been done by [Weber \(s.f\)](#)

3. Gauss method optimization

From the previous section, it can be inferred that knowing the observation times with certainty is essential to successfully determine the orbit of an asteroid, that is the reason that a code is executed, for organized date ranges in the observations.

We work with two sets of observations. The first has samples of observations for each NEA's (5587 (1990 SB) and 4953 (1990 MU)), randomly distributed over time, provided by the GAIA probe; For each data sample, the values with the same observation dates are eliminated and permutation combinations are made between the resulting data in order to obtain the largest amount of observational data. The second set consists of two samples of observations ordered in time (each one corresponding to each body), separated by days, starting with intervals from 1 to 264. It is worth mentioning that the choice of the two bodies for which their orbit, is carried out by virtue of the quantity and separation between dates of observations made by the GAIA probe and its classification.

After inferring the possible orbits for each body and taking into account each database, the data obtained are compared with those calculated by NASA's JPL (Jet Propulsion Laboratory) and compiled in the MPC (Minor Planet Center), it is important to mention that JPL works with the construction of a state vector that is obtained by means of radar

astrometry (Ryan Park (2022)), technology that is rarely available in many astronomical observatories around the world.

The contrast operation for the validation of the degree of confidence of the applied method is carried out by means of a linear fit on the orbital elements inferred for the GAIA data, the JPL horizons data and the orbital elements obtained for these bodies in the MPC. Linear regression analysis is performed using the least squares fitting method, with support from the Python Scipy library. The entire implementation, adaptation and data analysis procedure is developed in the Python programming language (†). The method is coded for each database in a Jupyter notebook in the github repository:: <https://gitfront.io/r/user-9182374/kmqWenqrt2CE/Congreso-IAU-2022/>.

On the other hand, through the numerical solutions incorporated in the NASA SPICE kernel in Python for a period of 100 years, with a step of 1 month, the degree of approximation of the elements inferred with the proposed method and the one used is verified. by JPL Horinzons, by means of the closeness to the curves that represent the variations of the orbital elements (a, e, i, ω, Ω) of each one of the bodies when subjected to perturbations, for this, the Sun-Earth-Moon-NEO system is modeled.

4. Results

The orbital elements are shown (delimited in a range of values according to the orbital characteristics of the NEA), the error calculated for the orbit obtained with each of the data sets and the one that meets the condition of being the most consistent with that determined by the JPL Horizons. For each of the bodies and with the results of each of the databases used, the first four sets of orbital element values are chosen. (a, e, i, ω, Ω) calculated, whose characteristic is approximately to the theoretical values determined by the MPC. Finally, the perturbed orbital elements are plotted, along with those inferred by each method. In total, 546 sets of observations (α, δ, t) are used for the first body, which is the love type, and 264 sets of observations (α, δ, t) for the second body, this being an apollo type under the category of potentially dangerous; the above data is for the GAIA probe.

At next, it is represented in histograms (figure 2 and figure 3), the distribution of points corresponding to the orbital elements calculated according to delta of randomly distributed observations for GAIA data and progressively systematized with deltas of increasing days with JPL Horizon. Note the number of calculated orbital element values that correspond and are close to those measured by the MPC for the epoch August 09, 2022 for both GAIA (left column) and JPL Horizon (right column) data.

From these represented data, for the orbits of the two minor bodies in question, the list of observations whose total orbital elements are closer to those measured by the MPC are filtered. In total, the best four orbits with the lowest standard deviations with respect to the values reported by the MPC are delimited. There are four orbits for each data set (GAIA, JPL) of each body. These new orbital element datasets are plotted as a linear regression, in order to estimate a global correlation coefficient between data measured by the MPC and those calculated for each data source (GAIA and JPL) for each body.

The expected value of said coefficient is 1, and corresponds to the correlation 1.1 for each of the MPC orbital elements and calculated according to each source. The ordering of the values is given automatically, and they are represented on a logarithmic scale. The correlation coefficients for each data set are arranged in the table 1.

Finally, from the four orbits selected for each body with each database, the one that is closest to the one calculated by the MPC is chosen. The results of the orbital elements that define the best orbit calculated with each database, are shown in the table 2.

† PythonSoftwareFoundation (2022). <https://www.python.org>

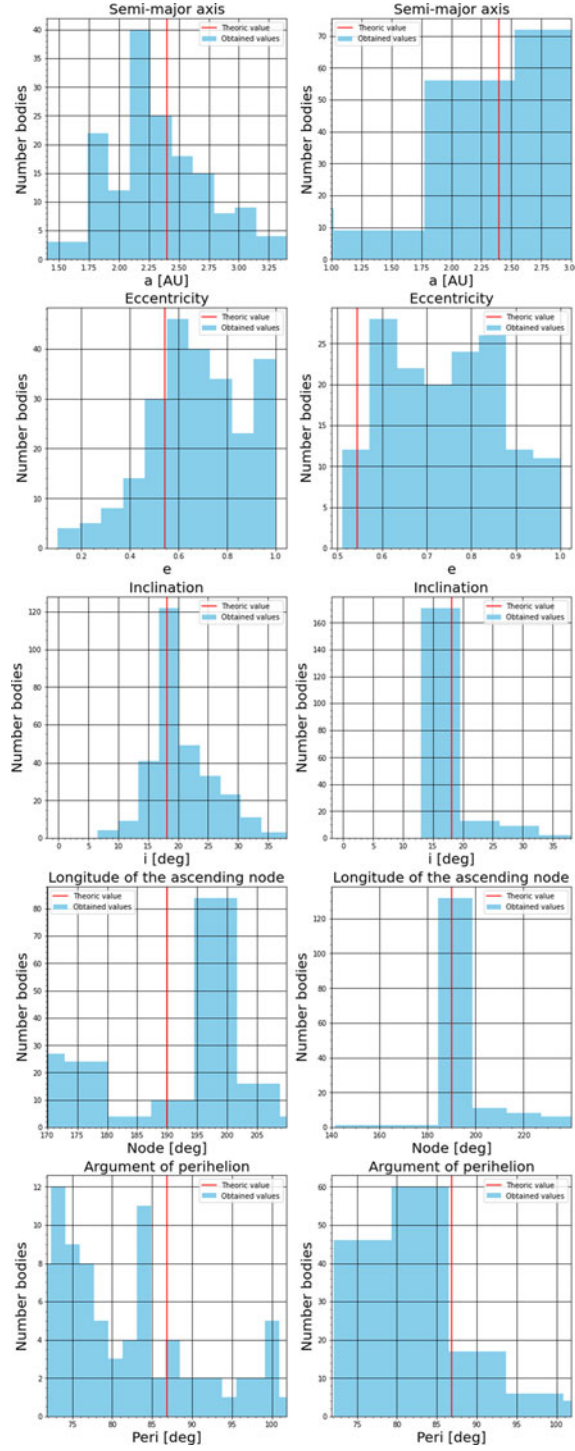


Figure 2. Histograms of orbital data for body 5587 (1990 SB), obtained with data from the GAIA probe and JPL Horizons.

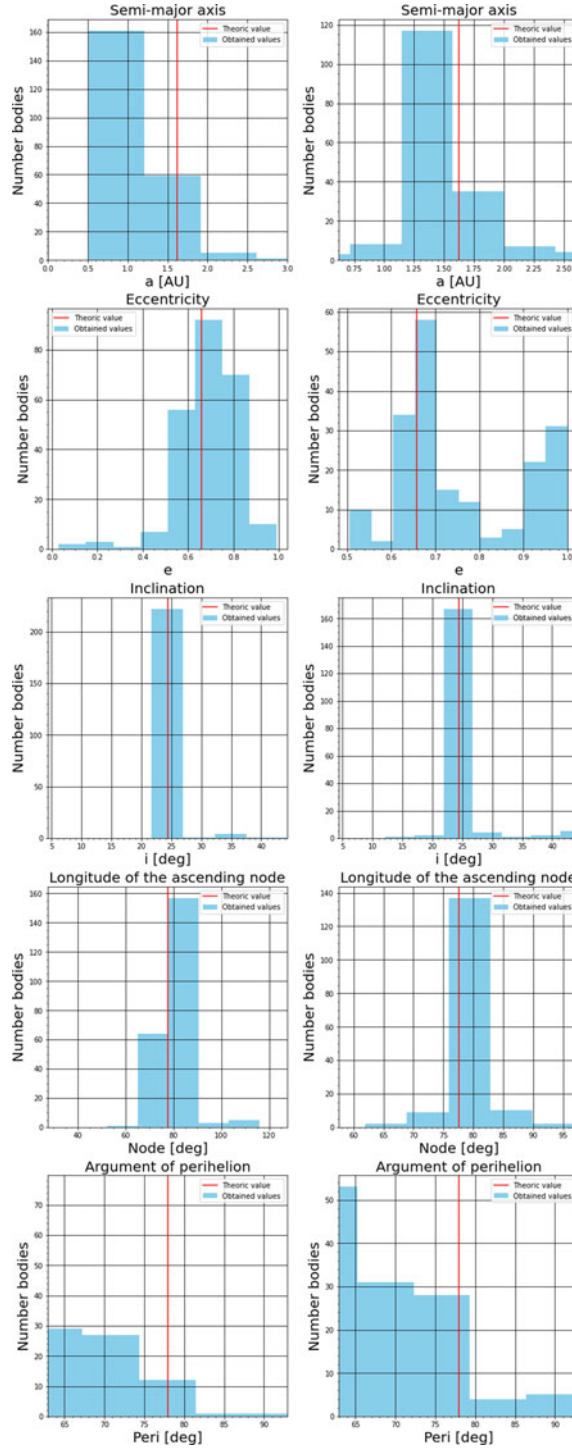


Figure 3. Histograms of orbital data for body 4953 (1990 MU), obtained with data from the GAIA probe and JPL Horizons.

Table 1. Correlation coefficients for bodies 5587 (1990 SB) and 4953 (1990 MU)

Name of NEA	Database	Slope	R^2
5587 (1990SB)	GAIA	0.852	0.906
5587 (1990SB)	JPL Horizons	0.993	0.997
4953 (1990 MU)	GAIA	0.997	0.999
4953 (1990 MU)	JPL Horizons	1.013	0.994

Table 2. Orbital elements obtained for the best orbit of the bodies 5587 (1990 SB) and 4953 (1990 MU) with each of the databases (JPL-Horizons and Gaia), compared to those determined by the MPC

Name of NEA	Database	a (AU)	e	i (deg)	Node (deg)	Peri (deg)
5587(1990 SB)	MPC	2.3976	0.5441	18.0946	189.9204	86.8275
5587(1990 SB)	JPL- Horizons	2.1953	0.5528	18.1252	196.1242	86.1973
5587(1990 SB)	Gaia	2.2858	0.5935	18.0893	196.1899	75.0662
4953 (1990 MU)	MPC	1.6216	0.6574	24.3739	77.5534	77.9210
4953 (1990 MU)	JPL- Horizons	1.3721	0.6512	24.3398	77.7148	77.9633
4953 (1990 MU)	Gaia	1.3922	0.6333	23.9818	73.4068	76.9167

Table 3. Error calculation of the computed orbits for bodies 5587 (1990 SB) and 4953 (1990 MU)

Name of NEA	Database	Error (%)
5587 (1990SB)	JPL Horizons	0.424149
5587 (1990SB)	GAIA	0.416237
4953 (1990 MU)	JPL Horizons	0.257968
4953 (1990 MU)	GAIA	0.223521

On the table 3, The error for each calculated orbit is presented, with respect to the measurements made by MPC for the year 2022.

In the context of the results previously obtained, a comparison was established between the orbital elements derived from the two NEOs studied, from the GAIA and JPL Horizons ephemeris databases, and the numerical solutions reported by the temporal evolution of the vector of state obtained using the NASA SPICE library, which incorporates second-order perturbations for the orbital elements of the bodies studied in an N-body astrodynamic problem. The program was executed for a time interval of 100 years with deltas corresponding to 1 month, and later the temporal evolution of the state vector of each body was transformed into functions that showed variability in their orbital elements. Contrasting the temporal evolution of each one of the orbital elements of the two studied NEOs, with the orbital elements inferred according to the two different data sources (GAIA and JPL Horizons), and the elements orbitals calculated for the corresponding epoch reported by the MPC through JPL, we obtained the figure 4.

It can be notice that, in general, the proposed optimization method produces orbital elements that are better approximated than those reported by MPC, while working with ephemeris data from JPL Horizons. “Best Approximation” refers to the optimization method takes values for each orbital element, which intersect with the values of the corresponding orbital elements, in the curves produced by numerical computation of state vectors in an N-body problem.

5. Conclusions

The Gauss method proposes that data collection should be equally spaced between observations, however, it is evident that although the GAIA data do not get this condition and are difficult to link to the method. It is possible to obtain approximate orbits to the

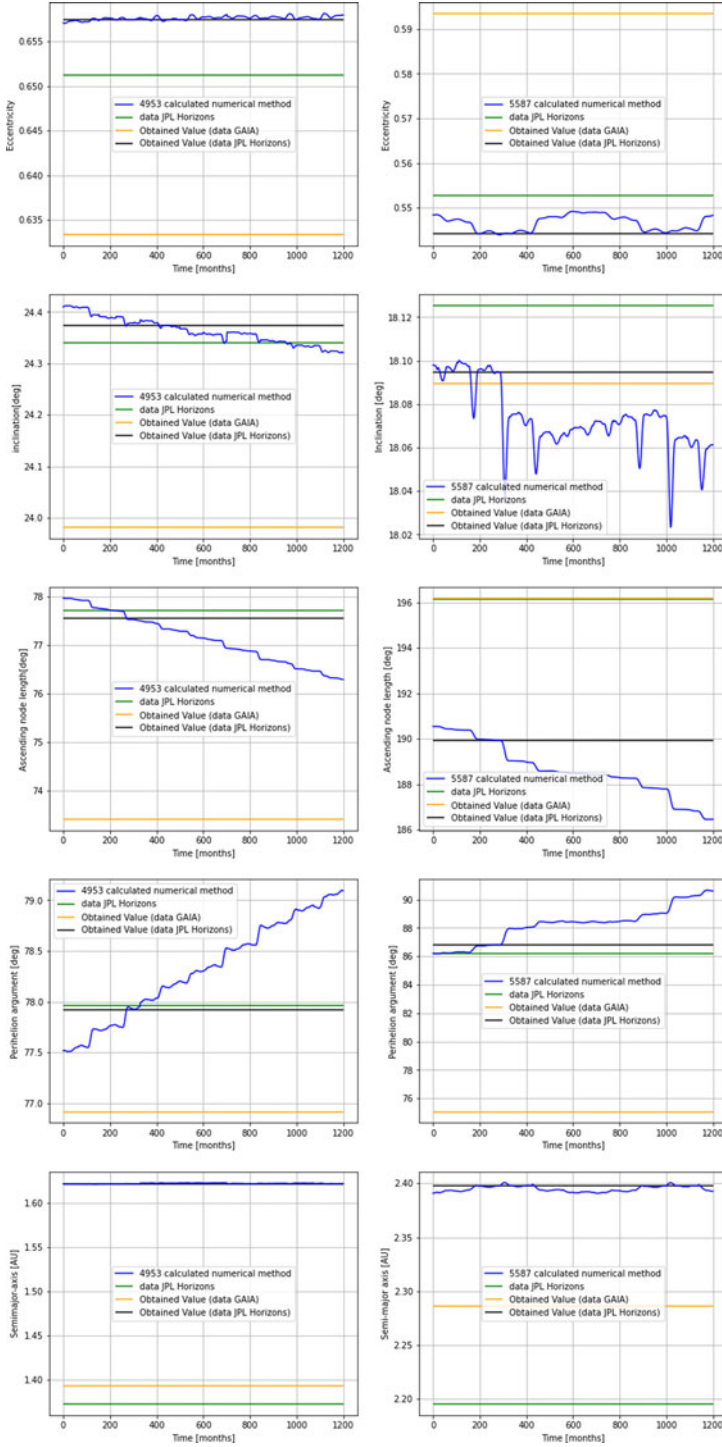


Figure 4. Temporal variability of orbital elements of the two analyzed NEOs. The data in the left column correspond to 4953 (1990 MU), and those in the right column to 5587 (1990 SB).

resulting ones with the JPL Horizons data and those calculated by the MPC, under the implementation of the corrections proposed in this study.

The proposed method in this work can be incorporated to optimize the observation in small observatories, since only 3 observations are needed without the necessity of radar technology or very high resolution telescopes, to calculate preliminary orbits of small bodies whose approximation is consistent with the data calculated by the MPC.

The obtained results for the calculation of the orbital elements through the optimization method incorporated here, in contrast to the results reported by the MPC for the last corresponding period of observation (August 2022), bring up the problem of establishing with respect to what sets of orbital elements we are evaluating the error in the inference of the orbits. The error results in the comparison of the different derivations, with what is established by the theoretical models that predict by means of numerical methods the way in which the orbital elements of some small body in the solar system varied in an N-body problem. If this is so, the question remains open about the criteria that prioritize theoretical derivations over empirical estimates made with observations both by MPC, as well as by our method, or any other that is incorporated in this direction.

Taking as reference the curves that produce the solution of the N-body orbital problem translated into the temporal evolution of each one of the orbital elements, our optimization method for JPL data suggests better approximate results than those reported by the MPC, and those calculated in our method with GAIA data.

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